Dilara Öztürk

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Education

Bilkent University, Faculty of Applied Sciences — September 2022 – June 2027 (expected date)

- Bachelor of Science Degree in Information Systems and Technologies
- GPA: 3.80/4.00

Metropolia University of Applied Sciences – Helsinki, Finland — August 2025 – December 2025

- Erasmus Exchange Program

İlke Birey Private Anatolian High School, Trabzon — September 2018 – June 2022

- High School Diploma
- GPA: 98.90/100.00

Awards and Achievements

- Maintained High Honor status (GPA: 3.90) and received Bilkent University's **Merit Scholarship** in recognition of academic excellence in 2024.

Experience

Software Development Intern, *Bilişim A.Ş. (ERP Development)* – Ankara, Türkiye — January 2026 – May 2026

- Built frontend and backend features on a Java-based ERP system.
- Improved GUI layouts, added business validations, and tightened data integrity.
- Migrated SOAP services to REST and tested them with Postman and Swagger.
- Worked with DAO architecture, SQL, Git, and enterprise development workflows using Oracle DB, Maven, and Tomcat.

Software Testing Intern, *TRT (Turkish Radio and Television Corporation)* – Ankara, Türkiye — June 2025 – July 2025

- Performed API and load testing using Postman, Insomnia, and k6 to ensure system reliability.
- Conducted end-to-end and UI automation testing with Cypress and Selenium (Java).
- Gained hands-on experience in software testing processes and quality management.

Projects

Personal Projects, *Music & Audio*

- Built **Guitar Splitter**, a local music source separation tool in **Python** using the Demucs neural network to isolate guitar, vocal, drum, bass, and piano stems from a song, with subtractive mixing to preserve original audio fidelity and lossless FLAC/MP3 export.
- Developed **Guitar Sound Effects**, a browser-based guitar pedal built with **React**, **TypeScript**, and the Web Audio API, featuring real-time distortion, a 3-band parametric EQ, live spectrum analysis, and multi-channel audio interface input.
- Ported the pedalboard to a native audio plugin, **SoundBox** (AU and VST3), in **C++** with **JUCE**: noise gate, compressor, 4×-oversampled distortion, 3-band EQ, impulse-response cabinet convolution, chorus, delay and reverb, with MIDI footswitch mapping and preset save/recall, running inside DAWs such as Logic Pro.

Course Projects, *Bilkent University*

- Developed a **Football Management System** in **Java** with team registration, match simulations, and finalist selection using structured algorithms.
- Recreated the **Flappy Bird** game in **C#**, implementing collision detection and dynamic gameplay.
- Built a **Racing Car Simulator** using **GLUT** and **OpenGL**, integrating collision detection and responsive controls.

- Developed a **Galaxy Shooter Game** with **GLUT** and **OpenGL**, enabling interactive spaceship control and projectile-based gameplay.
- Designed a responsive **Holiday Planner Website** using **HTML**, **CSS**, and **JavaScript** with an interactive user interface.
- Built a **MusicShop Marketplace Website** in **PHP**, featuring user authentication, database integration, and full frontend/backend synchronization.

Key Skills

- Collaborative team player with strong interpersonal and problem-solving skills.
- Skilled in identifying issues and developing innovative solutions.
- Experienced in managing multiple software projects and game development tasks effectively.
- Demonstrates leadership and organizational skills through active involvement in student organizations and team projects.
- Ability to quickly learn and adapt to new technologies and programming languages.

Programming Languages & Tools

- **Languages:** C, C++, Java, Python, JavaScript, TypeScript, SQL, PHP, HTML, CSS
- **Audio & Music Technology:** JUCE, Web Audio API, Demucs, Logic Pro
- **Testing & Automation Tools:** Postman, Insomnia, Selenium, Cypress, k6
- **Development & IDEs:** Visual Studio, Eclipse, Visual Studio Code, XCode
- **Databases:** MySQL, Oracle SQL Developer
- **Operating Systems & Platforms:** Windows, macOS, Linux, Parrot OS (terminal-based programming), Raspberry Pi
- **Other Tools:** Git, OpenGL, GLUT, JSON

Interests

Music production and audio software, programming, game development, animation, software testing, mobile applications and web systems.

Personal Interests and Hobbies: Music and playing instruments (guitar, violin, ukulele, piano) — YouTube Channel

Languages: Turkish, English.